

Konstantinos Kogkalidis
CURRICULUM VITAE

| **Structure-Aware Machine Learning** | **(AI for) Type & Proof Theory** |
| **Neurosymbolic Integration** | **Computational Linguistics** |

Updated July 6, 2026.

Personalia

Date of Birth 19 July 1991
Nationality Greek
Current Residence Thessaloniki, Greece
Pronouns he/him
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Positions

09/2025 - today Senior AI Specialist

Affiliation [Input Output \(IOG\)](#)
As part of the organization's AI transition, I advise on AI strategy & innovation policy and chair collaborative initiatives with academic partners. I also conduct applied research and own R&D initiatives in topics such as: (i) auto(in)formalization, autodocumentation & verifiable retrieval, (ii) cryptoeconomic auditing and adversarial security with reinforcement learning, (iii) syntactically-constrained neural code models for DSLs, (iv) LLM auditability and verifiability, and (v) neurosymbolic methods for agentic harnesses & self-improvement loops.

07/2023 - 07/2025 Postdoctoral Researcher

Affiliation Department of Computer Science, Aalto University
Conducted independent & collaborative research in neurosymbolic machine learning. Co-authored two publications at NeurIPS 2024 (main track) on the topics of representation learning and automated theorem proving. Awarded two independent personal grants for short-term research visits.

09/2024 - 02/2025 Visiting Researcher

Affiliation Department of Linguistics, University of Bologna
Research visit in the context of my Aalto post-doc. Co-authored and published a paper in CLiC-IT, the Italian computational linguistics conference.

02/2024 - 12/2024 ML Research Scientist

Affiliation [Skyfora](#)
Part-time role, concurrent with my post-doctorate, held remotely during my stay in Bologna. Developed and trained physics-informed neural models and ingestion pipelines for real-time, high-resolution, hyper-localized weather forecasting.

09/2018 - 12/2022 Doctoral Researcher

Affiliation Utrecht Institute of Linguistics, Utrecht University
Conducted research in neurosymbolic machine learning at the intersection of type theory, categorial grammars, and neural structure models. Adapted a modal λ -calculus to describe linguistic dependency and constituency structures. Engineered an extraction framework to translate parse graphs into syntactic proof-derivations, and produced a corresponding dataset for written Dutch. Designed neural parsing architectures to structurally enforce type-theoretic constraints during inference. Co-authored 11 papers, and (co-)taught 6 graduate course installments. For my thesis, I was awarded the FoLLIE.W. Beth Dissertation Prize. My first year as a PhD researcher was concurrent with the second year of my MSc studies.

07/2017 - 09/2018 ML EngineerAffiliation [Geekbot](#)

Part-time role, concurrent with the first year of my MSc studies, held remotely. Implemented in-house variants of then-current NLP models for unsupervised representation learning, topic modelling and semantic retrieval. Developed data sanitization and ingestion techniques to train and deploy at scale.

Education

09/2018 - 12/2022 PhD Candidate in *Computational Linguistics*

Institution Utrecht Institute of Linguistics, Utrecht University

Thesis [Dependency as Modality, Parsing as Permutation](#)

Promotors Michael Moortgat & Richard Moot

Committee A. v. d. Bosch, M. Sadrzadeh, J. Bos, S. Clark, S. Chatzikyriakidis

Award [E. W. Beth Dissertation Prize, FoLLI](#) (2024)**09/2017 - 07/2019 Research MSc** in *Artificial Intelligence* (120 EC, *cum laude*)

Track Logic & Reasoning

Institution Graduate School of Natural Sciences, Utrecht University

11/2010 - 05/2017 Engineering Diploma in *Electrical & Computer Engineering* (300 EC, MSc equivalent)

Track Electronics & Computers

Institution Department of Engineering, Aristotle University of Thessaloniki

Teaching

2022 - 2023 Compositional Models of Vector-Based Semantics (ESSLLI 2022, guest)**2021 - 2022** Logic & Language (UU AI MSc, guest, TA)**2020 - 2021** Logic & Language (UU AI MSc, main instructor)**2019 - 2020** Machine Learning for Human Vision & Language (UU AI MSc, TA)**2018 - 2019** Logic & Language (UU AI MSc, tutorials, TA)

Advanced Topics in Cognitive Modeling (UU AI MSc, TA)

Scholarships, Grants & Awards

12/2024 Concurrent Programming and Lambek Calculi (Working Visit)

Awarded Amount 1,500 €

Awarding Body Royal Netherlands Academy of Arts and Sciences ([KNAW](#))

Host Institution University of Groningen

Host Jorge Pérez

09/2024 Neural Premise Selection for Agda (Short-Term Scientific Mission)

Awarded Amount 1,330 €

Awarding Body European Research Network on Formal Proofs ([EuroProofNet](#))

Host Institution Chalmers

Host Jean-Philippe Bernardy

07/2024 E. W. Beth Dissertation Prize (Best Dissertation)

Awarded Amount 1,500 €

Awarding Body The Association for Logic, Language and Information ([FoLLI](#))

Publications & Preprints

PhD Thesis

[Dependency as Modality, Parsing as Permutation](#)

In *LOT Dissertation Series*. 2023.

Papers

1. [Nominal Class Assignment in Swahili: A Computational Account](#)
Giada Palmieri and Konstantinos Kogkalidis
In *Proceedings of the 10th Italian Conference on Computational Linguistics (CLiC-it 2024)*.
2. [On Tables with Numbers, with Numbers](#)
Konstantinos Kogkalidis and Stergios Chatzikyriakidis
In *Proceedings of the 1st Workshop on Language Models for Underserved Communities (LM4UC 2025)*.
3. [Learning Structure-Aware Representations of Dependent Types](#)
Konstantinos Kogkalidis, Orestis Melkonian and Jean-Philippe Bernardy
In *Proceedings of the 38th Conference on Neural Information Processing Systems (NeurIPS 2024)*.
4. [Algebraic Positional Encodings](#)
Konstantinos Kogkalidis, Jean-Philippe Bernardy and Vikas Garg
In *Proceedings of the 38th Conference on Neural Information Processing Systems (NeurIPS 2024)*.
5. [OYXOY: A Modern NLP Test Suite for Modern Greek](#)
Konstantinos Kogkalidis, Stergios Chatzikyriakidis, Eirini Chrysovalantou Giannikouri, Vassiliki Katsouli, Christina Kironomou, Christina Koula, Dimitris Papadakis, Thelka Pasparaki, Erofilis Psaltaki, Efthymia Sakellariou and Hara Soupiona
In *Findings of the Association for Computational Linguistics (EACL 2024)*.
6. [SPINDLE: Spinning Raw Text into Lambda Terms with Graph Attention](#)
Konstantinos Kogkalidis, Michael Moortgat and Richard Moot
In *Proceedings of the 17th Conference of the European Chapter of the Association for Computational Linguistics: System Demonstrations (EACL 2023 Demos)*.
7. [Geometry-Aware Supertagging with Heterogeneous Dynamic Convolutions](#)
Konstantinos Kogkalidis and Michael Moortgat
In *Proceedings of the 2023 CLASP Conference on Learning with Small Data (LSD 2023)*.
8. [Improving BERT Pretraining with Syntactic Supervision](#)
Georgios Tziafas, Konstantinos Kogkalidis, Gijs Wijnholds and Michael Moortgat
In *Proceedings of the 2023 CLASP Conference on Learning with Small Data (LSD 2023)*.
9. [Discontinuous Constituency and BERT: A Case Study of Dutch](#)
Konstantinos Kogkalidis and Gijs Wijnholds
In *Findings of the Association for Computational Linguistics (ACL 2023 Findings)*.
10. [Diamonds are Forever: Theoretical and Empirical Support for a Dependency-Enhanced Type Logic](#)
Michael Moortgat, Konstantinos Kogkalidis and Gijs Wijnholds
In *Logic and Algorithms in Computational Linguistics 2021 (LACompLing2021)*.
11. [A Logic-Based Framework for Natural Language Inference in Dutch](#)
Lasha Abzianizde and Konstantinos Kogkalidis
In *Journal of Computational Linguistics in the Netherlands (CLIN 2021)*.
12. [Fighting the COVID-19 Infodemic with a Holistic BERT Ensemble](#)
Georgios Tziafas, Konstantinos Kogkalidis and Tommaso Caselli
In *Proceedings of the 4th Workshop on NLP for Internet Freedom (NLP4IF 2021)*.
13. [Neural Proof Nets](#)
Konstantinos Kogkalidis, Michael Moortgat and Richard Moot
In *Proceedings of the 24th Conference on Computational Natural Language Learning (CoNLL 2020)*.

14. **[Æthel: Automatically Extracted Typelogical Derivations for Dutch](#)**
Konstantinos Kogkalidis, Michael Moortgat and Richard Moot
In *Proceedings of the 12th Language Resources and Evaluation Conference (LREC 2020)*.
15. **[Deductive Parsing with an Unbounded Type Lexicon](#)**
Konstantinos Kogkalidis, Michael Moortgat, Richard Moot and Georgios Tziafas
In *Preprint*. (2019).
16. **[Constructive Type-Logical Supertagging with Self-Attention Networks](#)**
Konstantinos Kogkalidis, Michael Moortgat and Tejaswini Deoskar
In *Proceedings of the 4th Workshop on Representation Learning for NLP (REPL4NLP 2019)*.
17. **[Towards a 2-Multiple Context-Free Grammar for the 3-Dimensional Dyck Language](#)**
Konstantinos Kogkalidis and Orestis Melkonian
In *At the Intersection of Language, Logic and Information (ESSLLI 2018 Student Session)*.

Presentations (selection)

1. [Invited] **Grammaticality as Provability**
At *The Logic Seminar of the University of Groningen*, Groningen, 2025.
2. [Contributed] **On Tables with Numbers, with numbers**
At *The NAACL 2025 Workshop on Language Models for Underserved Communities*, Online, 2025.
3. [Invited] **LLM-free Representation Learning from Theorem Structure**
At *Deep Learning Models for Mathematics and Type Theory*, Gothenburg, 2025.
4. [Contributed] **Structure-Aware Neural Representations of Agda Programs**
At *EuroProofNet-WG5: Theorem Proving and Machine Learning in the Age of LLMs*, Edinburgh, 2025.
5. [Contributed (Poster, Spotlight)] **Algebraic Positional Encodings**
At *Neural Information Processing Systems*, Vancouver, 2024.
6. [Contributed (Poster)] **Learning Structure-Aware Representations of Dependent Types**
At *Neural Information Processing Systems*, Vancouver, 2024.
7. [Invited] **Tables with Numbers, and What They (Do Not) Mean**
At *Computational Linguistics and Language Technology Lab (CLLT) at the University of Crete*, Online, 2024.
8. [Invited] **Empiricism in Computational Linguistics**
At *Circolo Linguistico dell'Università di Bologna*, Bologna, 2024.
9. [Contributed (Poster)] **Nominal Class Assignment in Swahili: A Computational Account**
At *Computational Linguistics in Italy*, Pisa, 2024.
10. [Contributed] **Geometry-Aware Supertagging with Heterogeneous Dynamic Convolutions**
At *The 2023 CLASP Conference on Learning with Small Data*, Gothenburg, 2023.
11. [Contributed (Poster)] **Improving BERT Pretraining with Syntactic Supervision**
At *The 2023 CLASP Conference on Learning with Small Data*, Gothenburg, 2023.
12. [Contributed] **Neurosymbolic Proof Search for Linguistics**
At *Logic for the AI Spring, Lake Como School of Advanced Studies*, Como, 2023.
13. [Invited] **The Unicorn of Constant-Time Parsing**
At *End-to-End Compositional Models of Vector-Based Semantics, ESSLLI*, Galway, 2022.
14. [Lecture] **Logic-Based Parsing with Graph Neural Networks**
At *End-to-End Compositional Models of Vector-Based Semantics, ESSLLI*, Galway, 2022.
15. [Contributed] **Supertagging Beyond Trees with Heterogeneous Dynamic Convolutions**
At *Computational Linguistics in the Netherlands*, Tilburg, 2022.
16. [Internal (Demo)] **From Text to Meaning: A Type-Driven Treebank, and Applications**
At *Language Sciences Day*, Utrecht, 2021.
17. [Contributed] **Neural Proof Nets**
At *The 24th Conference on Computational Natural Language Learning*, Online, 2020.

18. [Contributed] **Deductive Parsing with an Unbounded Type Lexicon**
At *SemSpace 4, ESSLLI*, Riga, 2019.
19. [Contributed (Poster)] **Constructive Type-Logical Supertagging with Self-Attention Networks**
At *The 4th Workshop on Representation Learning for NLP*, Florence, 2019.
20. [Contributed] **From Raw Text to Linear Lambda Terms**
At *Compositionality in Formal and Distributional Models of Natural Language Semantics*, Utrecht, 2019.
21. [Contributed] **Towards a 2-MCFG for the 3-Dimensional Dyck Language**
At *ESSLLI Student Session*, Sofia, 2018.

Professional Service

ACL ARR

Reviewer (2021 -); Area Chair (2025 -)

NeurIPS

Reviewer (2025 -)

Extracurricular

Languages

Greek [native], English [fluent], Italian [intermediate], German [elementary], Dutch [elementary]